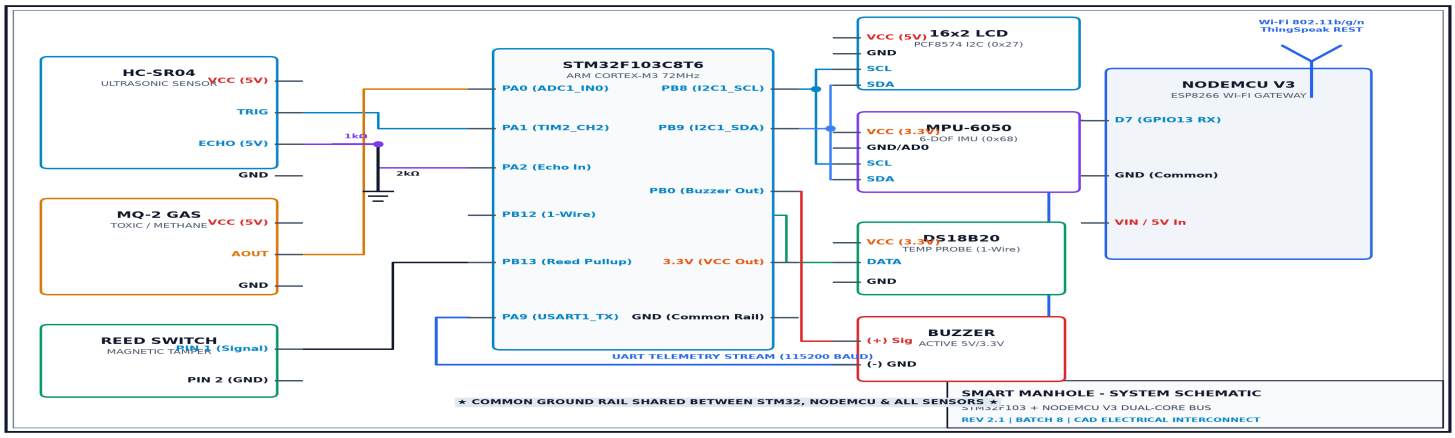
SMART MANHOLE UTILITIES URBAN IOT

Autonomous Sub-Surface Monitoring, Real-Time Cloud Telemetry & 3D Digital Twin SCADA

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1. PROFESSIONAL CIRCUIT SCHEMATIC & HARDWARE INTERCONNECT

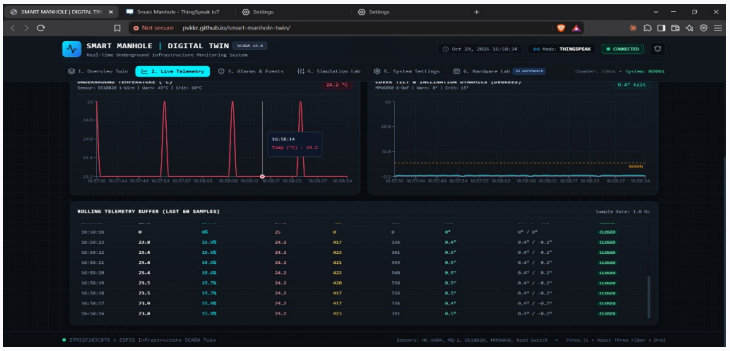


Electrical Interconnect CAD Blueprint: Bare-Metal STM32F103 Master, NodeMCU ESP8266 Gateway, 6-Sensor Bus & 1kΩ/2kΩ Voltage Divider

2. PHYSICAL HARDWARE PROTOTYPE & 3D DIGITAL TWIN SCADA INTERFACE

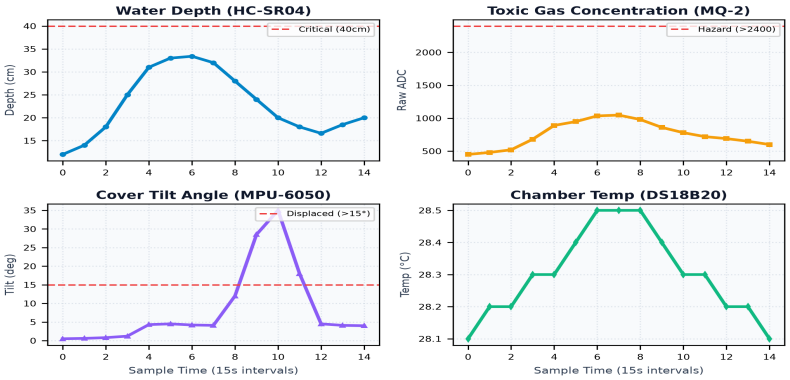


Benchtop Hardware Circuit: 16x2 LCD + NodeMCU + STM32



Live 3D Digital Twin: Physics-accurate water level & lid tilt

3. MULTI-SENSOR LIVE TELEMETRY & FIELD PERFORMANCE



Real-Time Multi-Sensor Telemetry Graphs (ThingSpeak Channel #3558043)

Key Project Capabilities:

- **Flood Sensing:** HC-SR04 continuous water depth (2–400cm).
- **Worker Safety:** MQ-2 toxic gas alarm threshold (ADC >2400).
- **Cover Anti-Theft:** MPU-6050 3-axis tilt + Reed switch seal.

Engineering Bottlenecks Solved:

- **ARM Float Fix:** Nano.specs float stripped; resolved with %d.%d.
- **I2C Lockup:** Eliminated bus freeze with strict I2C_Stop & settle.
- **Zero-Latency:** Sub-20ms local REST (/telemetry) + 1s ThingSpeak.

Hardware & Interfacing:

- **Dual Core:** Bare-metal STM32 10Hz loop + NodeMCU ESP8266.
- **Voltage Divider:** 1kΩ/2kΩ passive divider protects PA2 pin.